

Large-Angle Statistics of the CMB Angular Correlation Function: Testing Λ CDM at Low Multipoles

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Abstract

We investigate the statistical properties of the cosmic microwave background (CMB) angular correlation function at large angular scales ($\ell \leq 30$) using Planck legacy data. Three complementary statistics are studied: the integrated squared correlation $S_{\theta_a}^{\theta_b}$, the interval-averaged correlation $\langle \xi \rangle_{\theta_a}^{\theta_b}$, and the antipodal correlation C_{180} . Theoretical distributions are generated via two independent methods, sampling from the MCMC posterior of the standard flat Λ CDM model fitted with the CamSpec high-multipole TT likelihood, and Monte Carlo realizations around the best-fit spectrum, allowing a direct comparison of the two approaches. We attribute the lack of large-angle correlations to a ‘look-elsewhere’ effect, as the discrepancy exists but the overall significance is reduced when considering multiple correlated statistics simultaneously.

Keywords: cosmic microwave background, angular correlation function, large-angle anomalies, Planck, Λ CDM

1. Introduction

Since the first measurements of the CMB temperature anisotropies, the largest angular scales have attracted particular attention because they probe the longest wavelength perturbations accessible within the observable universe. Early analyses of the COBE-DMR data already suggested that the angular two-point correlation function was unexpectedly close to zero for separations larger than approximately 60° (Hinshaw et al., 1996). Although the statistical power of COBE was limited, this feature was subsequently confirmed with much higher precision by the first-year WMAP observations (Spergel et al., 2003).

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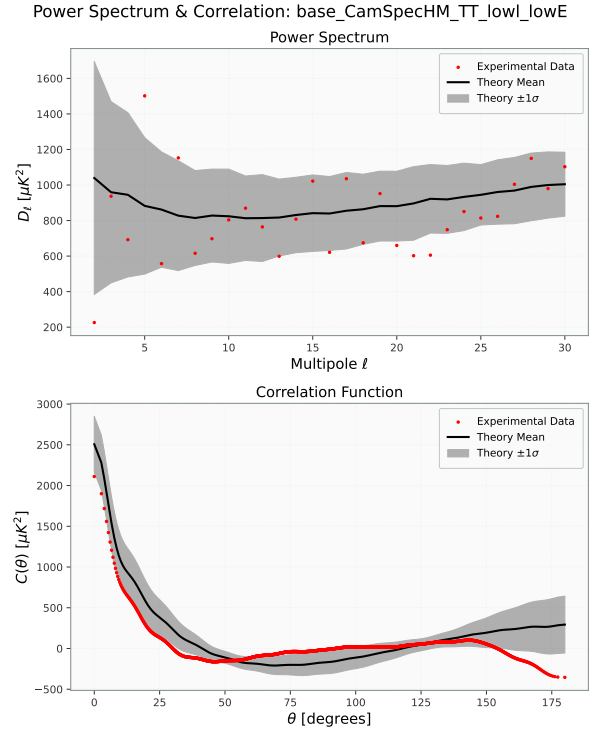


Figure 1: Power spectrum (top) and angular correlation function (bottom) for model `base_CamSpecHM_TT_low1_lowE`. Experimental Planck data (red points), theoretical mean (black line), and $\pm 1\sigma$ band (grey) are shown. The suppression of $C(\theta)$ for $\theta \gtrsim 60^\circ$ relative to the theoretical expectation is visible in the lower panel.

To quantify the apparent suppression of correlations on large angular scales, Spergel et al. introduced the statistic

$$S_{1/2} = \int_{-1}^{1/2} C(\theta)^2, d(\cos \theta),$$

which measures the integrated squared correlation function over angular separations larger than 60° (Spergel et al., 2003). The observed value of $S_{1/2}$ was found to lie in the extreme tail of the Λ CDM distribution generated from Monte Carlo realizations, establishing the lack of large-angle correlations as one of the most persistent large-scale anomalies in the CMB.

A series of studies by Copi, Huterer, Schwarz and Starkman strengthened this conclusion using successive WMAP data releases. In particular, analyses of cut-sky maps demonstrated that the observed correlation function is nearly zero for $\theta \gtrsim 60^\circ$, with probabilities typically below 10^{-3} under the standard cosmological model (Copi et al., 2006, 2009, 2013). These works emphasized that the anomaly cannot be explained simply by the low quadrupole amplitude alone, but instead reflects an unusual covariance structure among the lowest multipoles. They also noted connections with other large-scale anomalies, including the quadrupole-octopole alignment and hemispherical asymmetries.

The interpretation of the anomaly, however, remains controversial. Efstathiou, Ma and Hanson argued that estimators based directly on masked-sky correlation functions can exaggerate the discrepancy with Λ CDM, and showed that full-sky maximum-likelihood reconstructions yield results more consistent with theoretical expectations (Efstathiou et al., 2010). Extending this argument, Pontzen and Peiris considered a broader class of anisotropic Gaussian models and concluded that the observed correlation function does not constitute compelling evidence for new physics once the full space of possible covariance structures is taken into account (Pontzen and Peiris, 2010). These studies highlighted the importance of estimator choice, a posteriori statistic selection, and the definition of the theoretical ensemble used when assessing significance.

Another line of investigation has focused on the relationship between the lack of large-angle correlations and other low-multipole anomalies. Sarkar

et al. tested whether the suppression of $C(\theta)$ is statistically connected to the well-known quadrupole-octopole alignments and found essentially no correlation between the two effects within Λ CDM realizations (Sarkar et al., 2010). The simultaneous occurrence of both anomalies is therefore substantially rarer than either anomaly considered individually. Similarly, Kim and Naselsky argued that the observed odd-parity preference at low multipoles provides a phenomenological explanation for the deficit of large-angle correlations, although no convincing physical origin was identified (Kim and Naselsky, 2011).

The anomaly has also motivated the development of alternative statistical descriptions of large-scale correlations. Direction-dependent generalizations of the correlation function were introduced to investigate whether the missing power is localized on the sky rather than uniformly distributed. Zhang constructed directional versions of both $C(\theta)$ and $S_{1/2}$, showing that a large fraction of the full-sky signal originates from regions associated with the quadrupole and octopole extrema (Zhang, 2012). Other approaches include bipolar harmonic decompositions (Joshi et al., 2009), generalized isotropic angular correlation functions (Dipanshu et al., 2023), geometric representations of the two-point function (Froes et al., 2015), and all-angle statistics such as S_1 , proposed to reduce the dependence on the a posteriori choice of the 60° threshold (Gruppuso, 2014).

The persistence of the anomaly in increasingly precise datasets has further motivated investigations beyond temperature anisotropies. Analyses of Planck polarization maps have revealed a similar suppression of large-angle correlations, with significances comparable to those found in temperature data (Chiocchetta et al., 2020). If confirmed, such results would suggest that the anomaly is not solely a consequence of temperature-specific systematics and may instead reflect a more fundamental property of the largest observable scales.

Despite nearly two decades of study, there is still no consensus regarding either the statistical significance or the physical interpretation of the lack of large-angle correlations. Part of this ambiguity arises from the existence of many partially correlated test statistics, each emphasizing different aspects of the low- ℓ temperature field. A rigorous assessment therefore requires both a careful construc-

tion of the theoretical sampling distribution and a framework capable of combining evidence from multiple correlated observables.

2. Statistics

The statistics studied here are designed to probe complementary aspects of the large-angle correlation function: its integrated power, its mean level over specific angular ranges, and its value at antipodal separation. The choice is guided by the existing literature on low- ℓ anomalies, though as noted by Efstathiou et al. (2010), some statistics, including $S_{1/2}$, were identified a posteriori from the data, which inflates their apparent significance. Studying multiple statistics simultaneously, and combining their p-values through a global test, is therefore essential to obtain a reliable assessment of tension with Λ CDM.

2.1. Angular Power Spectrum and Correlation Function

The CMB temperature anisotropy $\Delta T(\hat{n})/T_0$ is expanded in spherical harmonics with coefficients $a_{\ell m}$. For a statistically isotropic Gaussian field the two-point statistics are fully encoded in the angular power spectrum $C_\ell = \langle |a_{\ell m}|^2 \rangle$. We work with the conventional scaled spectrum $D_\ell = \ell(\ell+1)C_\ell/(2\pi)$, which is approximately flat for a scale-invariant primordial spectrum.

The angular two-point correlation function is obtained by transforming back to real space via a Legendre expansion,

$$C(\theta) = \sum_{\ell} \frac{2\ell+1}{4\pi} C_{\ell} P_{\ell}(\cos \theta) = \sum_{\ell} \frac{2\ell+1}{2\ell(\ell+1)} D_{\ell} P_{\ell}(\cos \theta), \quad (1)$$

where P_{ℓ} are Legendre polynomials. The sum runs from $\ell = 2$ to $\ell_{\max} = 30$ throughout this work.

2.2. Interval Statistics

Three statistics are evaluated on each of the angular intervals listed in Table~1, chosen to partition the accessible range $[0^\circ, 180^\circ]$ into physically motivated regions while including the classic $S_{1/2}$ interval for comparison with the literature.

Antipodal correlation $C_{180} \equiv C(\theta = 180^\circ)$ measures the correlation between antipodal points

on the sky. Because it is a pointwise evaluation of $C(\theta)$ it is relatively insensitive to masking effects at intermediate scales, and it provides a direct test of the near-vanishing of large-angle correlations at the largest possible separation.

Interval-averaged correlation $\langle \xi \rangle_{\theta_a}^{\theta_b}$ is the mean of $C(\theta)$ over the angular interval $[\theta_a, \theta_b]$, expressed in terms of $\mu = \cos \theta$ as

$$\langle \xi \rangle_{\theta_a}^{\theta_b} = \frac{1}{b-a} \int_a^b C(\theta) d\mu = \sum_{\ell} \frac{D_{\ell}}{2\ell(\ell+1)} \frac{P_{\ell+1}(b) - P_{\ell-1}(b) - P_{\ell+1}(a) + P_{\ell-1}(a)}{b-a}, \quad (2)$$

where $a = \cos \theta_b$ and $b = \cos \theta_a$. Being linear in D_{ℓ} , this statistic is sensitive to shifts in the mean correlation level within an interval and changes sign when the correlation function crosses zero, making it complementary to $S_{\theta_a}^{\theta_b}$.

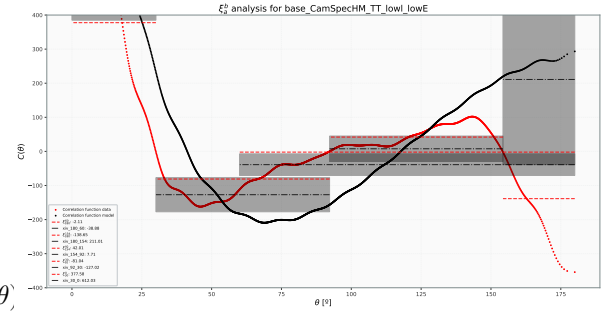


Figure 2: Interval-averaged correlation function for model `base_CamSpecHM_TT_low1_lowE` MCMC chains

Integrated squared correlation $S_{\theta_a}^{\theta_b}$ is defined as

$$S_{\theta_a}^{\theta_b} = \int_a^b C^2(\theta) \sin \theta d\theta = \sum_{n,m} A_n A_m D_n D_m T_{nm}, \quad (3)$$

Table 1: Angular intervals used for the statistics.

Angular range	$[\cos \theta_b, \cos \theta_a]$
$[0^\circ, 30^\circ]$	$[0.865, 1.000]$
$[30^\circ, 92^\circ]$	$[-0.039, 0.865]$
$[92^\circ, 154^\circ]$	$[-0.901, -0.039]$
$[154^\circ, 180^\circ]$	$[-1.000, -0.901]$
$[60^\circ, 180^\circ]$	$[-1.000, 0.500]$

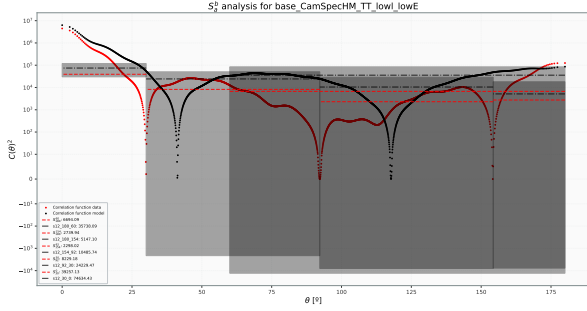


Figure 3: Integrated squared correlation function for model `base_CamSpecHM_TT_lowl_lowE` MCMC chains

where $A_\ell = (2\ell + 1)/[2\ell(\ell + 1)]$ and

$$T_{nm} = \sum_{r=0}^{\min(m,n)} \frac{A_r A_{m-r} A_{n-r} [P_{m+n-2r+1} - P_{m+n-2r-1}]_a^b}{A_{m+n-r} (2m + 2n - 2r + 1)}$$

are overlap integrals of Legendre polynomials over the interval (Carlitz, 1961) similar expression to the one obtained in (Copi et al., 2009), but extended to different angles. Being quadratic in D_ℓ , this statistic amplifies deviations that are coherent across multipoles and is always non-negative, so it cannot be cancelled by sign changes of $C(\theta)$ within the interval. The classic $S_{1/2}$ statistic of Spergel et al. (2003) corresponds to the choice $[\theta_a, \theta_b] = [60^\circ, 180^\circ]$.

The intervals considered are listed in Table~1, expressed both in terms of angular separation and the corresponding cosine values. These intervals were chosen based on the zeros of the correlation function derived from the power spectrum obtained by (Aghanim et al., 2020) rather than fixed angular thresholds and we included the already measured $S_{1/2}$.

3. Data

3.1. Planck Posterior Chains

We use the publicly available Planck 2018 MCMC chains from the Planck Legacy Archive¹ (Planck Collaboration, 2015). The primary dataset is the CamSpec high-multipole TT likelihood combined with the low- ℓ temperature and polarization likelihoods (`base_CamSpecHM_TT_lowl_lowE`). This provides posterior samples of the six standard Λ CDM parameters $\{\Omega_b h^2, \Omega_c h^2, H_0, \tau, n_s, A_s\}$ along with a suite of derived quantities. The Plik-based TT likelihood and two curvature-extended runs are considered in Appendix~B.

For each posterior sample the theoretical power spectrum D_ℓ is computed with CAMB (Lewis), and the corresponding $C(\theta)$ and derived statistics are evaluated analytically using equations 1–3.

3.2. Observed Planck Multipoles

The experimental values of each statistic are computed from the Planck 2018 Commander temperature map and associated beam and pixel window functions, using the same multipole range $\ell \in [2, 30]$. These observed values serve as the reference against which the theoretical distributions are compared.

3.3. Ensemble Construction

Two independent strategies are used to construct the theoretical sampling distribution.²

1. **MCMC ensemble.** For each of the 1000 samples drawn from the tail of the Planck posterior chain, specifically the last 1000 rows after GetDist’s default burn-in removal, the cosmological parameters are passed to CAMB to compute a theoretical spectrum D_ℓ^{th} . Each spectrum is then passed through the simulation pipeline described below, producing one realisation of cosmic variance on top of the parameter-determined D_ℓ^{th} . The resulting ensemble therefore captures both parameter uncertainty and cosmic variance simultaneously, constituting a Bayesian predictive distribution for each statistic.

¹<https://pla.esac.esa.int/>

²The full analysis pipeline is publicly available at <https://github.com/PhyAMR/CMB>.

2. Best-fit ensemble. The best-fit D_ℓ^{bf} is read from the `.minimum.theory_c1` file of the Planck chain, which contains the spectrum evaluated at the maximum-posterior parameter point. This single spectrum is used as input to 1000 independent pipeline realisations, each with a different random seed. The spread of the resulting ensemble is driven entirely by cosmic variance at fixed cosmological parameters, with no contribution from parameter uncertainty. The width of its histograms is therefore a lower bound on the total variance: the additional width seen in the MCMC histograms is attributable purely to parameter uncertainty, and comparing the two ensembles provides a clean decomposition of the two contributions.

In both cases, the pipeline follows the methodology of [Sullivan et al. \(2024\)](#). For each theoretical C_ℓ^{th} spectrum, the beam transfer function b_ℓ and the HEALPix pixel window function p_ℓ are applied, a full-sky temperature map is synthesised at $N_{\text{side}} = 2048$ using `hp.synfast`, and the pseudo- C_ℓ are recovered with `hp.anafast`. The recovered spectrum is then deconvolved by $b_\ell^2 p_\ell^2$ to obtain an estimate $\hat{D}_\ell$ in the same space as the published Planck values. When a sky mask is applied, the recovered C_ℓ are corrected by the sky fraction f_{sky} before deconvolution. At $\ell_{\text{max}} = 30$ the beam and pixel window corrections are numerically negligible, but they are included for consistency with the Planck data reduction.

In order to reduce the look-elsewhere effect and obtain a calibrated global significance that accounts for the correlation structure of the statistics without requiring an explicit model of it, we use the Aggregated Cauchy Association Test (ACAT; [Liu and Xie, 2020](#)). The eleven statistics examined, five intervals each for $S_{\theta_a}^{\theta_b}$ and $\langle \xi \rangle_{\theta_a}^{\theta_b}$, plus C_{180} , are not independent: all derive from the same realisation of D_ℓ , so their p-values are correlated by construction. Treating them as independent and applying a simple Bonferroni correction would be overly conservative, while reporting only the most extreme individual p-value would be anti-conservative.

For each statistic $k \in \{1, \dots, 11\}$ the individual p-value p_k is computed as the one-sided empirical quantile of the observed value within the simulated ensemble: $p_k = F_k$, where F_k is the fraction of realisations with a statistic value less than or equal to

the observed one. The ACAT test statistic is then

$$T = \sum_{k=1}^{11} w_k \tan\left[\left(\frac{1}{2} - p_k\right) \pi\right],$$

where equal weights $w_k = 1/11$ are used, reflecting no a priori preference for any particular angular interval or statistic. Under the null hypothesis, T follows a standard Cauchy distribution regardless of the dependence structure among the p_k , so the combined p-value is obtained analytically as

$$p_{\text{global}} = \frac{1}{2} - \frac{\arctan T}{\pi}.$$

The Cauchy combination is exact under arbitrary dependence, including the strong correlations expected between statistics computed over overlapping angular ranges. The global tension in units of Gaussian standard deviations is reported as $n_\sigma = \Phi^{-1}(1 - p_{\text{global}}/2)$, where Φ^{-1} is the probit function.

4. Results

4.1. Best-Fit Analysis

In the best-fit analysis the theoretical sampling distribution is generated by Monte Carlo realizations of the CMB sky drawn from the best-fit Λ CDM spectrum. This approach isolates cosmic variance from parameter uncertainty.

Figures [A.4](#) and [A.5](#) display the distributions of the $S_{\theta_a}^{\theta_b}$ and $\langle \xi \rangle_{\theta_a}^{\theta_b}$ statistics respectively, for all five angular intervals. Each panel shows the distribution over the 1000 Monte Carlo realizations (blue histogram), and the observed Planck value (red vertical line). Figure [A.6](#) shows the analogous distribution for C_{180} .

As we can see, for the different intervals all values of $S_{\theta_a}^{\theta_b}$ are below the median of the distribution, which is not only consistent with the lack of large-angle correlations but also suggests that the lack is not only present at large angles but also at intermediate and small angles. Nevertheless, the observed values are not in the extreme tails of the distributions, and the empirical p-values are not low enough to claim a significant tension with Λ CDM.

In this set of histograms we can see that the observed values of $\langle \xi \rangle_{\theta_a}^{\theta_b}$ sometimes are lower and sometimes higher than the median of the distribution,

which is consistent with the fact that $\langle \xi \rangle_{\theta_a}^{\theta_b}$ is a linear statistic, and therefore it is more sensitive to shifts in the mean correlation level, and the change of sign in the correlation function in certain intervals.

The statistical summary, including the median, 68% credible interval, observed value, and empirical p-value for each statistic, is reported in Table~2.

4.2. MCMC Analysis

In the MCMC analysis the theoretical sampling distribution is constructed directly from the Planck posterior, propagating full parameter uncertainty through to the derived statistics. The 1000 posterior samples drawn from the CamSpec Λ CDM chain each yield a distinct D_ℓ spectrum, whose distribution reflects both cosmic variance and parameter degeneracies.

Figures A.7, A.8, and A.9 show the resulting distributions of $S_{\theta_a}^{\theta_b}$, $\langle \xi \rangle_{\theta_a}^{\theta_b}$, and C_{180} respectively. The broader width of the MCMC-derived histograms compared to the best-fit case reflects the additional spread due to cosmological parameter uncertainty.

The quantitative results for the MCMC ensemble are collected in Table~3.

5. Discussion

The results for alternative likelihoods and models with free curvature Ω_k are reported in Appendix~B. The dependence of the results on the likelihood choice (CamSpec vs. Plik) provides a useful cross-check, while the curvature-extended models test whether a departure from flatness could alleviate the observed tensions.

As noted in (Efstathiou et al., 2010), in the low- ℓ regime, the theoretical distribution of the statistics is mostly independent of the mask used, and the effects of the mask are negligible.

As noted by (Kim and Naselsky, 2011), a lack of correlation is observed across all angular scales—including large, intermediate, and small angles. This is reflected in the fact that the observed values of $S_{\theta_a}^{\theta_b}$ fall below the distribution’s median for all intervals. Consequently, the absence of large-angle correlation extends to smaller scales, where the observed values of $\langle \xi \rangle_{\theta_a}^{\theta_b}$ fluctuate above and below the median. Because $\langle \xi \rangle_{\theta_a}^{\theta_b}$ is a linear statistic,

it is naturally more sensitive to overall shifts in the mean correlation level as well as sign changes within specific intervals.

6. Conclusions

We have studied the large-angle CMB angular correlation function and its derived statistics using Planck 2018 data and two complementary ensembles for the theoretical sampling distributions. The statistics examined, the quadratic $S_{\theta_a}^{\theta_b}$, the linear $\langle \xi \rangle_{\theta_a}^{\theta_b}$, and the antipodal C_{180} — probe distinct aspects of the large-angle temperature field.

Both ensembles provide complementary perspectives on the significance of the large-angle anomalies. The best-fit ensemble isolates the contribution of cosmic variance, while the MCMC ensemble captures the full parameter uncertainty and the cosmic variance.

A complete summary of all p-values and multivariate tension estimates for the primary `base_CamSpecHM_TT_lowl_lowE` model is provided in Tables~2 and 3; results for additional models appear in Appendix~B.

While the lack of large-angle correlations is a persistent feature of the observed CMB sky, specially the C_{180} statistic, the Λ CDM model is not in significant tension with the data when the full distribution of statistics is taken into account. The observed values are generally below the median of the theoretical distributions, consistent with the lack of large-angle correlations.

Another raised question from previous analyses is whether the lack of large-angle correlations is a statistical fluke or a hint of new physics. The observed p-values for the C_{180} statistic are low across multiple simulations and models, and while they are in the order of $2-2.5\sigma$ the fact that it is persistent across all our simulations makes it more clear that the lack of large-angle correlations is a real feature of the data, and not a statistical fluke.

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Table 2: Statistical summary for the best-fit analysis of model `base_CamSpecHM_TT_low1_lowE`. Columns give the experimental value, best-fit median, 68% interval, and empirical p-value.

Global tension (ACAT): $T = 2.091$, $p = 0.142$ (1.07σ)				
Statistic	Experimental	Median	68% CI	p-value
C_{180}	-354.1497	260.7976	[-38.2572, 629.7045]	0.022 (2.01σ)
$S_{180,60}$	6694.0882	35725.0265	[11308.2691, 108785.4640]	0.073 (1.45σ)
$S_{180,154}$	2739.9435	5144.2773	[704.4897, 26362.7403]	0.358 (0.36σ)
$S_{154,92}$	2298.0174	10463.2839	[2893.9033, 39807.8958]	0.133 (1.11σ)
$S_{92,30}$	8229.1759	24227.7116	[9406.5535, 62090.4792]	0.127 (1.14σ)
$S_{30,0}$	39257.1300	74487.4419	[42884.9866, 131879.2429]	0.114 (1.21σ)
$\xi_{180,60}$	-2.1052	-38.8758	[-78.4081, -14.6241]	0.966 (-1.83σ)
$\xi_{180,154}$	-138.6536	210.8664	[7.7966, 508.4919]	0.051 (1.64σ)
$\xi_{154,92}$	42.0072	7.6378	[-25.2907, 49.3801]	0.797 (-0.83σ)
$\xi_{92,30}$	-81.0404	-127.2322	[-184.4908, -84.2408]	0.862 (-1.09σ)
$\xi_{30,0}$	377.5806	611.8525	[424.7966, 878.9550]	0.098 (1.29σ)

Table 3: Statistical summary for the MCMC analysis of model `base_CamSpecHM_TT_low1_lowE`. Columns give the experimental value, MCMC median, 68% credible interval, and empirical p-value.

Global tension (ACAT): $T = 2.348$, $p = 0.128$ (1.14σ)				
Statistic	Experimental	Median	68% CI	p-value
C_{180}	-354.1497	261.6952	[-39.0156, 589.9108]	0.018 (2.10σ)
$S_{180,60}$	6694.0882	33937.6161	[10094.0365, 102450.9702]	0.076 (1.43σ)
$S_{180,154}$	2739.9435	5200.7872	[664.4140, 23292.6452]	0.368 (0.34σ)
$S_{154,92}$	2298.0174	10088.0224	[2572.3400, 34735.6347]	0.138 (1.09σ)
$S_{92,30}$	8229.1759	23088.7680	[9244.2759, 60281.6008]	0.137 (1.09σ)
$S_{30,0}$	39257.1300	72409.8155	[41887.8294, 128755.4170]	0.133 (1.11σ)
$\xi_{180,60}$	-2.1052	-37.1448	[-77.0816, -13.6870]	0.962 (-1.77σ)
$\xi_{180,154}$	-138.6536	210.0608	[-7.9407, 473.6089]	0.047 (1.67σ)
$\xi_{154,92}$	42.0072	8.7805	[-22.1704, 48.5205]	0.802 (-0.85σ)
$\xi_{92,30}$	-81.0404	-126.3688	[-182.5317, -81.9472]	0.844 (-1.01σ)
$\xi_{30,0}$	377.5806	603.5009	[414.7549, 861.6303]	0.116 (1.20σ)

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Appendix A. Monte Carlo Histograms

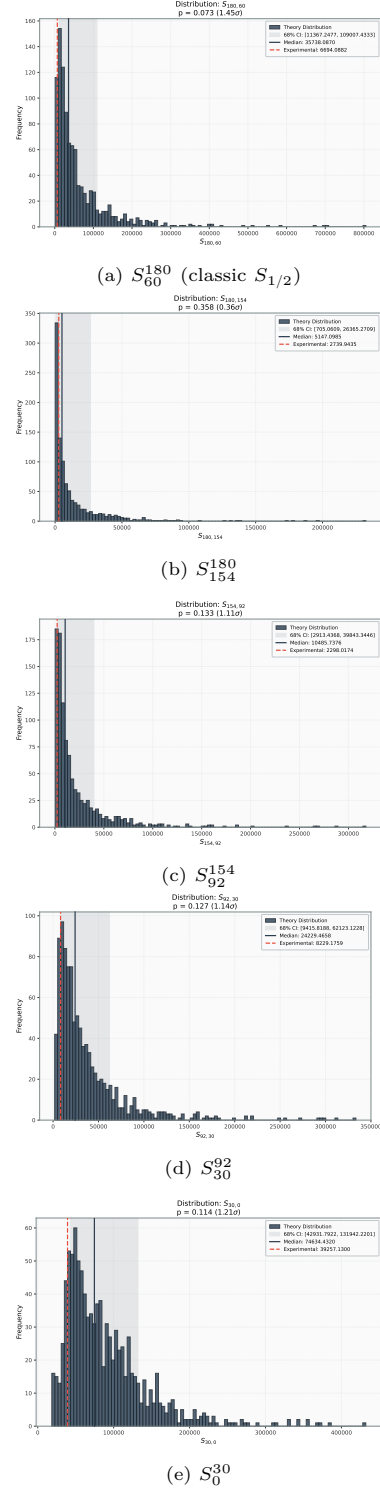
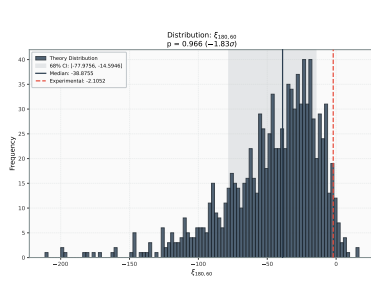
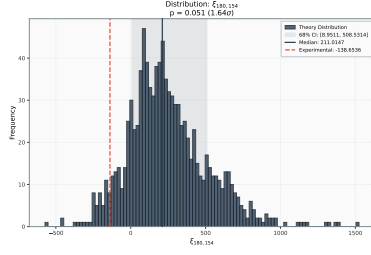


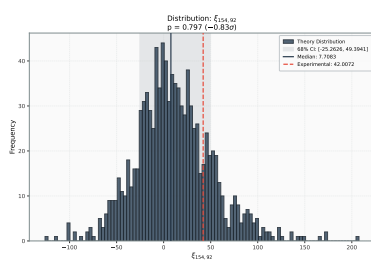
Figure A.4: Best-fit Monte Carlo distributions of the S_a^b statistic for model `base_CamSpecHM_TT_lowl_lowE` ($\ell_{\max} = 30$). Each panel shows the normalized histogram of 1000 realizations; the red line marks the observed Planck value. The x-axis is in logarithmic scale.



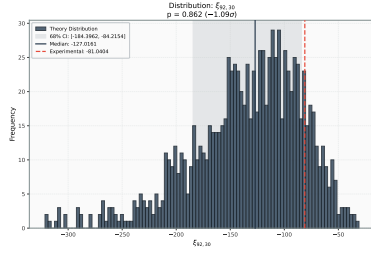
(a) $\langle \xi \rangle_{60}^{180}$



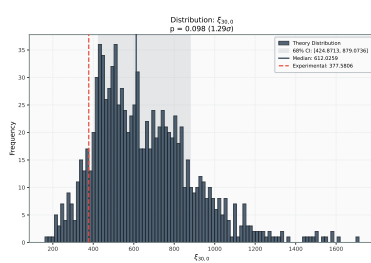
(b) $\langle \xi \rangle_{154}^{180}$



(c) $\langle \xi \rangle_{92}^{154}$



(d) $\langle \xi \rangle_{30}^{92}$



(e) $\langle \xi \rangle_{0}^{30}$

Figure A.5: Best-fit Monte Carlo distributions of the interval-averaged correlation $\langle \xi \rangle_{\theta_a}^{\theta_b}$ for model base_CamSpecHM_TT_low1_lowE.

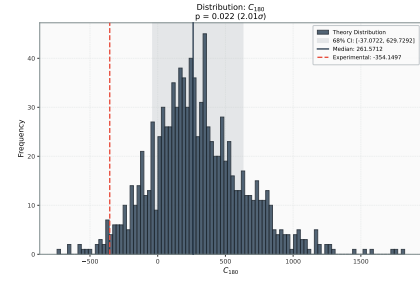
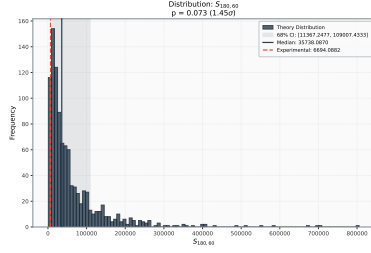
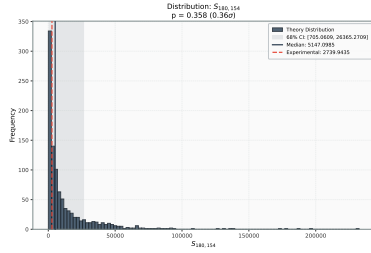


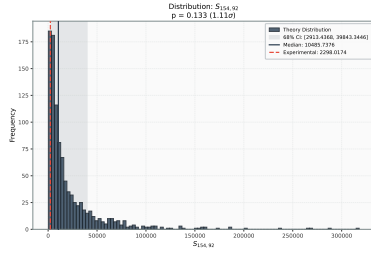
Figure A.6: Best-fit Monte Carlo distribution of the antipodal correlation C_{180} for model base_CamSpecHM_TT_low1_lowE.



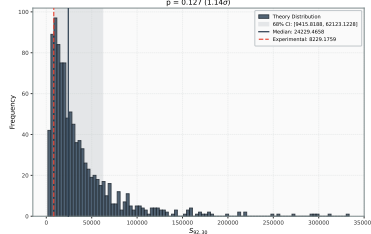
(a) S_{60}^{180} (classic $S_{1/2}$)



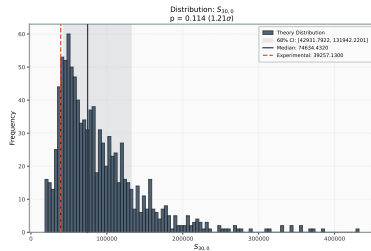
(b) S_{154}^{180}



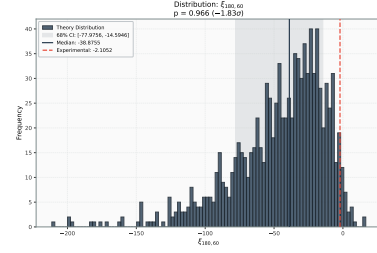
(c) S_{92}^{154}



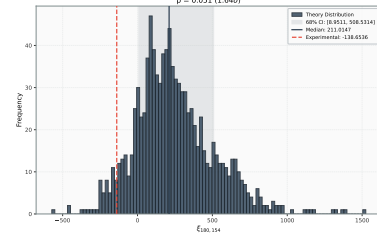
(d) S_{30}^{92}



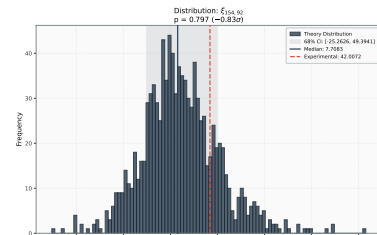
(e) S_0^{30}



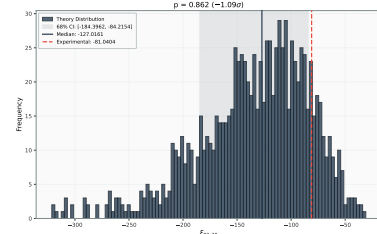
(a) $\langle \xi \rangle_{60}^{180}$



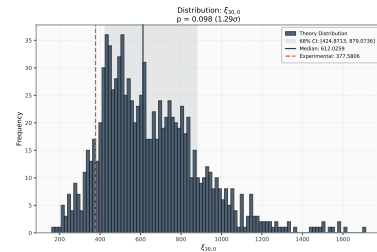
(b) $\langle \xi \rangle_{154}^{180}$



(c) $\langle \xi \rangle_{92}^{154}$



(d) $\langle \xi \rangle_{30}^{92}$



(e) $\langle \xi \rangle_0^{30}$

Figure A.7: MCMC posterior distributions of the $S_{\theta_a}^{\theta_b}$ statistic for model `base_CamSpecHM_TT_low1_lowE`. Each panel shows the distribution over 1000 posterior samples from the Planck CamSpec chain.

Figure A.8: MCMC posterior distributions of $\langle \xi \rangle_{\theta_a}^{\theta_b}$ for model `base_CamSpecHM_TT_low1_lowE`.

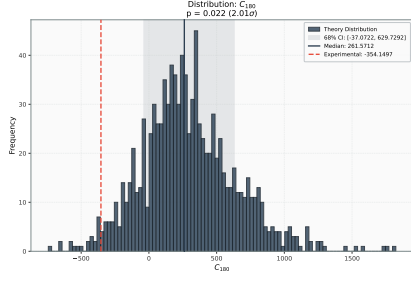


Figure A.9: MCMC posterior distribution of C_{180} for model `base_CamSpecHM_TT_lowl_lowE`.

Appendix B. Additional Models: Statistical Tables

The tables in this appendix report the full statistical summary for three additional Planck models. All quantities are defined as in the main text. Model names follow the Planck convention: `omegak` denotes a run with free spatial curvature; `plikHM` denotes the Plik high-multipole likelihood; `TTTEEE` denotes a joint temperature and polarization fit.

Appendix B.1. MCMC Results

Table B.4: MCMC results for `base_plikHM_TT_lowl_lowE`.

Global tension (ACAT): $T = 1.884$, $p = 0.155$ (1.01σ)				
Statistic	Experimental	Median	68% CI	p -value
C_{180}	-354.1497	241.2170	[-55.1699, 594.4383]	0.029 (1.90σ)
$S_{180,60}$	6694.0882	33801.8449	[11110.7265, 102182.3499]	0.065 (1.51σ)
$S_{180,154}$	2739.9435	4766.0801	[672.8630, 22097.9814]	0.376 (0.32σ)
$S_{154,92}$	2298.0174	10527.3578	[2848.7506, 34955.3646]	0.125 (1.15σ)
$S_{92,30}$	8229.1759	23310.8042	[9398.8164, 59071.8598]	0.128 (1.14σ)
$S_{30,0}$	39257.1300	73229.9679	[42537.9015, 125384.8849]	0.126 (1.15σ)
$\xi_{180,60}$	-2.1052	-38.5341	[-76.0073, -14.4676]	0.962 (-1.77σ)
$\xi_{180,154}$	-138.6536	202.3885	[-6.9888, 468.8826]	0.052 (1.63σ)
$\xi_{154,92}$	42.0072	9.1633	[-25.4628, 51.1022]	0.781 (-0.78σ)
$\xi_{92,30}$	-81.0404	-126.3941	[-178.6658, -84.8224]	0.860 (-1.08σ)
$\xi_{30,0}$	377.5806	607.6992	[416.0714, 851.9444]	0.092 (1.33σ)

Table B.5: MCMC results for `base_omegak_CamSpecHM_TT_lowl_lowE`.

Global tension (ACAT): $T = 3.250$, $p = 0.095$ (1.31σ)				
Statistic	Experimental	Median	68% CI	p -value
C_{180}	-354.1497	208.9942	[-34.4807, 486.2964]	0.011 (2.29σ)
$S_{180,60}$	6694.0882	21765.8299	[8109.5135, 74119.3517]	0.119 (1.18σ)
$S_{180,154}$	2739.9435	3267.8897	[391.6000, 15964.8461]	0.466 (0.09σ)
$S_{154,92}$	2298.0174	7021.0369	[2079.4052, 26823.1748]	0.186 (0.89σ)
$S_{92,30}$	8229.1759	15837.8991	[6161.9693, 41989.8915]	0.246 (0.69σ)
$S_{30,0}$	39257.1300	51438.9870	[29716.1148, 94216.3869]	0.310 (0.50σ)
$\xi_{180,60}$	-2.1052	-30.7119	[-64.0166, -12.3069]	0.957 (-1.72σ)
$\xi_{180,154}$	-138.6536	164.2943	[-0.5697, 396.1609]	0.030 (1.88σ)
$\xi_{154,92}$	42.0072	6.7193	[-20.7478, 41.0697]	0.847 (-1.02σ)
$\xi_{92,30}$	-81.0404	-105.5266	[-152.8439, -67.5677]	0.734 (-0.62σ)
$\xi_{30,0}$	377.5806	502.4040	[340.1959, 725.3792]	0.232 (0.73σ)

Table B.6: MCMC results for `base_omegak_plikHM_TTTEEE_lowl_lowE`.

Global tension (ACAT): $T = 2.615$, $p = 0.116$ (1.19σ)				
Statistic	Experimental	Median	68% CI	p -value
C_{180}	-354.1497	208.6258	[-45.5817, 506.9834]	0.013 (2.23σ)
$S_{180,60}$	6694.0882	25836.5779	[8014.3153, 74173.0515]	0.111 (1.22σ)
$S_{180,154}$	2739.9435	3617.6754	[464.5482, 16839.1825]	0.455 (0.11σ)
$S_{154,92}$	2298.0174	7418.2542	[2223.0737, 24954.1641]	0.172 (0.95σ)
$S_{92,30}$	8229.1759	18231.4590	[6845.5271, 42367.6853]	0.205 (0.82σ)
$S_{30,0}$	39257.1300	55436.6264	[32260.9180, 96229.5173]	0.264 (0.63σ)
$\xi_{180,60}$	-2.1052	-33.4438	[-63.2413, -11.6939]	0.961 (-1.76σ)
$\xi_{180,154}$	-138.6536	168.6679	[-15.5933, 403.9776]	0.040 (1.75σ)
$\xi_{154,92}$	42.0072	9.8408	[-20.0036, 45.5296]	0.816 (-0.90σ)
$\xi_{92,30}$	-81.0404	-110.8947	[-155.6151, -70.9814]	0.766 (-0.73σ)
$\xi_{30,0}$	377.5806	522.1450	[353.3445, 736.3491]	0.196 (0.86σ)

Appendix B.2. Best-Fit Results

Table B.7: Best-fit results for `base_plikHM_TT_lowl_lowE`.

Global tension (ACAT): $T = 1.753$, $p = 0.165$ (0.97σ)				
Statistic	Experimental	Median	68% CI	p -value
C_{180}	−354.1497	235.3880	[−62.2566, 586.2191]	0.028 (1.91σ)
$S_{180,60}$	6694.0882	33261.4972	[10608.9403, 100001.9703]	0.082 (1.39σ)
$S_{180,154}$	2739.9435	4638.8774	[579.8045, 21964.4118]	0.379 (0.31σ)
$S_{154,92}$	2298.0174	9746.7383	[2727.4077, 35795.7064]	0.130 (1.13σ)
$S_{92,30}$	8229.1759	23100.4791	[9000.3718, 60298.5831]	0.145 (1.06σ)
$S_{30,0}$	39257.1300	70442.7090	[40918.0369, 130962.8091]	0.137 (1.09σ)
$\xi_{180,60}$	−2.1052	−37.0548	[−75.1144, −13.0243]	0.959 ($−1.74\sigma$)
$\xi_{180,154}$	−138.6536	195.4772	[−25.0376, 460.2276]	0.061 (1.55σ)
$\xi_{154,92}$	42.0072	10.7736	[−22.3436, 56.5905]	0.774 ($−0.75\sigma$)
$\xi_{92,30}$	−81.0404	−124.4639	[−185.5235, −80.1492]	0.835 ($−0.97\sigma$)
$\xi_{30,0}$	377.5806	595.2820	[409.3589, 866.2165]	0.113 (1.21σ)

Table B.8: Best-fit results for `base_omegak_CamSpecHM_TT_lowl_lowE`.

Global tension (ACAT): $T = 2.686$, $p = 0.113$ (1.21σ)				
Statistic	Experimental	Median	68% CI	p -value
C_{180}	−354.1497	211.9556	[−29.8411, 502.2334]	0.014 (2.20σ)
$S_{180,60}$	6694.0882	23719.2671	[7675.5717, 73031.1322]	0.131 (1.12σ)
$S_{180,154}$	2739.9435	3625.1233	[406.7749, 15623.4577]	0.453 (0.12σ)
$S_{154,92}$	2298.0174	7128.0973	[1943.9653, 24772.1131]	0.209 (0.81σ)
$S_{92,30}$	8229.1759	16340.3031	[6726.9125, 41856.6421]	0.223 (0.76σ)
$S_{30,0}$	39257.1300	51705.4424	[31338.3396, 91825.7121]	0.295 (0.54σ)
$\xi_{180,60}$	−2.1052	−32.3840	[−63.4182, −11.6872]	0.953 ($−1.67\sigma$)
$\xi_{180,154}$	−138.6536	172.3367	[−0.9287, 394.2349]	0.034 (1.83σ)
$\xi_{154,92}$	42.0072	8.3009	[−20.7864, 42.4205]	0.835 ($−0.97\sigma$)
$\xi_{92,30}$	−81.0404	−105.6599	[−151.8632, −71.0518]	0.747 ($−0.67\sigma$)
$\xi_{30,0}$	377.5806	504.0592	[354.5585, 720.7156]	0.211 (0.80σ)

Table B.9: Best-fit results for `base_omegak_plikHM_TTTEEE_lowl_lowE`.

Global tension (ACAT): $T = 5.786$, $p = 0.054$ (1.60σ)				
Statistic	Experimental	Median	68% CI	p -value
C_{180}	−354.1497	218.3127	[−17.0392, 508.2853]	0.006 (2.51σ)
$S_{180,60}$	6694.0882	24349.6721	[7615.2117, 72133.8665]	0.136 (1.10σ)
$S_{180,154}$	2739.9435	3289.8516	[501.4208, 16347.5457]	0.459 (0.10σ)
$S_{154,92}$	2298.0174	7095.9810	[2120.8351, 26428.7002]	0.171 (0.95σ)
$S_{92,30}$	8229.1759	17049.6488	[6261.6264, 42388.9107]	0.224 (0.76σ)
$S_{30,0}$	39257.1300	54447.2715	[31086.6010, 96125.6258]	0.280 (0.58σ)
$\xi_{180,60}$	−2.1052	−33.5933	[−63.5443, −11.2915]	0.967 ($−1.84\sigma$)
$\xi_{180,154}$	−138.6536	172.7077	[9.0572, 400.9675]	0.020 (2.05σ)
$\xi_{154,92}$	42.0072	7.2581	[−21.0722, 40.1496]	0.853 ($−1.05\sigma$)
$\xi_{92,30}$	−81.0404	−106.5318	[−154.6407, −69.6381]	0.744 ($−0.66\sigma$)
$\xi_{30,0}$	377.5806	518.2218	[349.5821, 732.1247]	0.211 (0.80σ)